MASTER IN ASTROPHYSICS AND SPACE SCIENCE
УНИВЕРЗИТЕТ У БЕОГРАДУ
UNIVERSITY OF BELGRADE

Nuclear Starburst Disks

Active Galactic Nuclei

Fritz Ali Agildere

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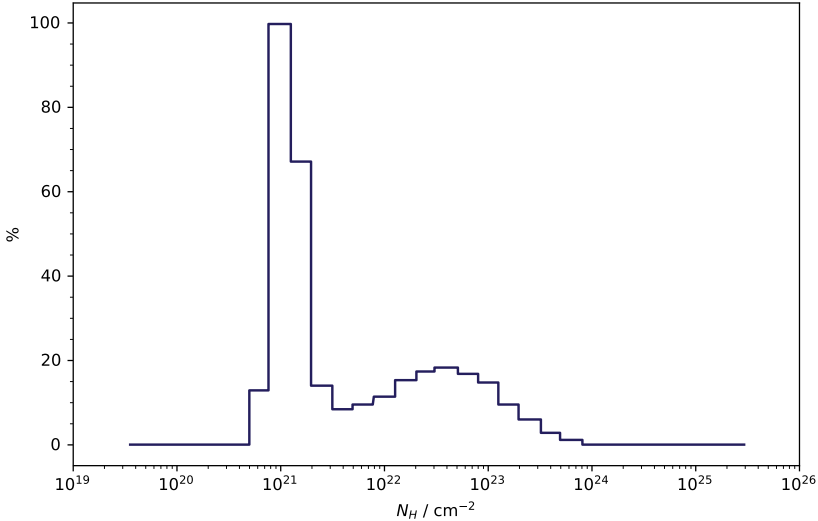
Motivation

- Natural Central SMBH Rapid Feeding Consequence
- Potentially Galaxy Merger Triggered
- Vertical Gas & Dust Expansion Intercepts Sightlines
- Fuel Region / Galactic Disk Kinematic Decoupling
- Spatial SF & AGN Luminosity Correlation
  - Galactic Scales $R \sim 10\,\mathrm{kpc}$ Weak
  - Compact Nucleus $R \sim 100\,\mathrm{pc}$ Strong
- Dynamically Evolving Oscuring Starforming Structure

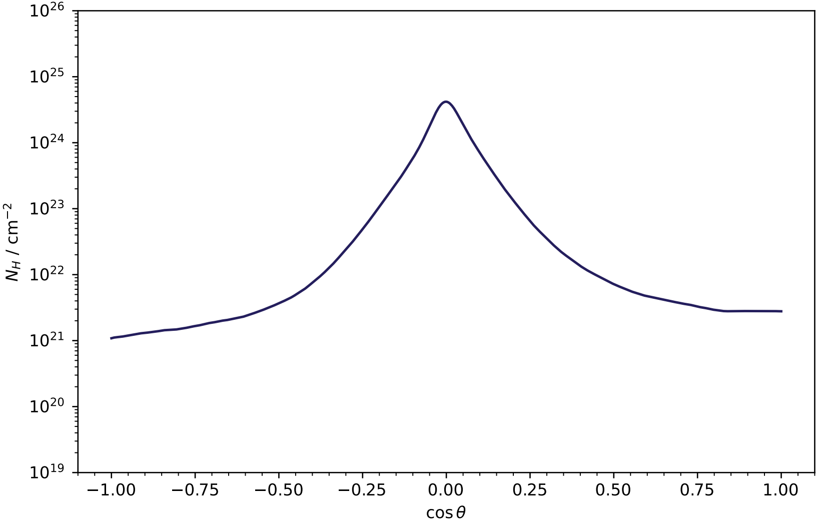
Structure

- Radiatively Supported Large Scale Heights (Star Formation)
    - Intense Stellar Radiation Pressure
    - Supernovae Kinetic Feedback
- Distinct Peripheral $R > 10\,\mathrm{pc}$ Regions (Compton Thin)
- Highly Localized $R < 1\,\mathrm{pc}$ Obscuration (Compton Thick)
- Embedded Stellar Remnant Thermal Effects
- Statistically Consistent Viewing Angle Sampled Distributions

Column Density



Angular Obscuring



Outlook

- Observational Static Torus / Puffy Compact Starburst Disk Separation
- Evolutionary Stage / Static Mechanism Distinction
- Refined Simulation Feedback Interactions (FIRE & STARFORGE)
- CO SLED Molecular Gas Tracers (ALMA)
- IR Starforming & Disk Signatures (JWST)

Thank You!

- R. C. Hickox & D. M. Alexander, *Obscured Active Galactic Nuclei* (2018) ARA&A Volume 56
- P. F. Hopkins et. al., *Stellar and Quasar Feedback in Concert: Effects on AGN Accretion, Obscuration and Outflows* (2016) MNRAS Volume 458 Issue 1

Resources on GitHub: here